Aurevia Master Plan: Prototype to Sellable Product

Oct 15, 2026 · @Virat

Purpose and status

This plan takes Aurevia from a working prototype to a product a company will pay for instead of a 10 or 20 person sales team. It is written for Claude Code: every phase is split into work packages (WP) with a goal, tasks and a "done when" test.

The product: a multi-tenant AI voice agent that cold calls prospects, qualifies them, handles objections, books meetings and follows up. It must feel like a friendly human call, say it is an AI when asked, and cost clearly less than the team it replaces.

Code written (Phases 1 to 9), not yet proven on real calls: multi-tenant backend with Postgres row-level security, voice worker (LiveKit, Deepgram, Cartesia), Gemini and Anthropic behind a Model Gateway, 12 audited sales tools, sales-stage machine, memory across calls, compliance gate with an India policy pack, campaigns and queue, usage and plans, a Next.js dashboard, and about 330 Python tests. This code exists and its tests pass, but it has never carried a real phone call. Speed, the human feel, learning from recordings, integrations and phone go-live are still missing or unproven, and the phases below cover them.

Not yet true (this plan closes these gaps):

- Replies take about 2.6 to 2.8 s at the median and 6.5 s at the slow end. The target is under 1 s.
- No backchannels ("hmm", "okay", "yaa") from the agent, and no emotion in the voice.
- The expert-call learning pipeline is design only.
- Cost is about 3 to 5 rupees per talk minute (estimate). The target is under 2.
- Phone calling is untested: the Exotel trunk is unconnected, there is no hosting, and the India policy has no lawyer sign-off.
- Bookings are internal records only. There is no calendar, CRM, WhatsApp, SMS or email.
- The voice has never passed a human microphone test, and no pilot has run.

How Claude Code works on this plan

Start every session by reading `CLAUDE.md`, `docs/AGENTS.md` and `docs/INDEX.md`. Those rules stay in force; this plan adds work on top of them.

Rules that never change

1. One work package (WP) per pull request. Never merge or deploy without being told. Never enable auto-merge.
2. Explain first, then act. If a WP needs an architecture change, stop, explain, propose and wait for approval.
3. Report honestly: which checks ran and their real results. Separate mocked tests from tests against a real provider, and a synthetic audio test from a human microphone test. Never claim a real call or booking that was only mocked.
4. Never ask for or print an API key. Secrets live only in `backend/.env` or the cloud secret settings.
5. The LLM is not the controller. Side effects only through validated, tenant-scoped, audited tools.
6. Every new tenant-owned table gets `tenant_id`, an index starting with it, and a forced row-level-security policy. Never edit an applied migration.
7. No real outbound call without the compliance gate. Live mode stays blocked until a lawyer approves the policy pack for that country. The auto-dialer stays off unless switched on in both the server setting and the campaign.
8. The agent never claims to be human and answers honestly when asked if it is an AI. Naturalness comes from timing, warmth and listening, never from pretending.
9. Provider SDKs stay behind adapters (`backend/src/aurevia/providers/`, `voice-worker/.../speech_providers.py`). Provider choice is configuration only.
10. Keep scope tight. Do not build a later phase early. Fix failing tests properly; never weaken a safety rule to turn CI green.

Every work package follows this template

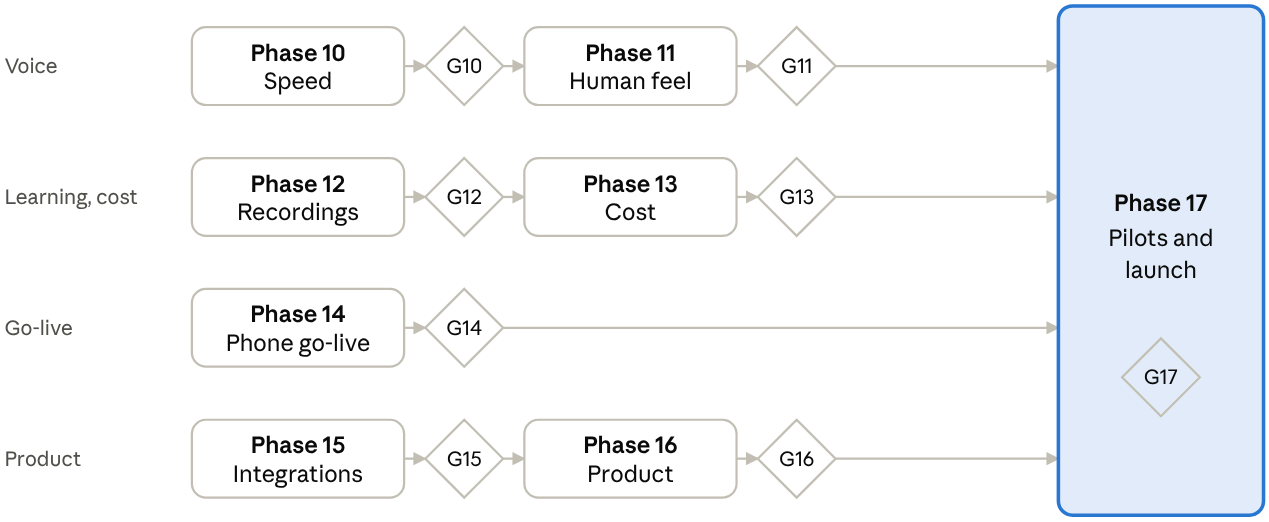
- **Goal:** the outcome in one sentence.
- **Tasks:** concrete steps, with the files or modules to touch.
- **Done when:** measurable checks, including the tests to add.
- **Report:** what ran, what passed, what was mocked, what is blocked and on whom.

Definition of done for any WP: backend `ruff`, `mypy` and `pytest` pass; voice-worker checks pass; dashboard typecheck, lint, test and build pass; the OpenAPI file and generated types are regenerated when an API changed; docs updated; no secrets in logs; PR opened and reported, not merged.

Roadmap at a glance

Four tracks run side by side and meet at the pilots. Each diamond is a gate that must pass before its phase counts as done.

Four tracks run side by side and meet at the pilots



Phase 18 (later) starts only after Gate 17.

roadmap · 4 tracks, 8 gates

Within a track the order shown is the recommended one, but work across tracks can overlap. Phase 12's timing profile feeds Phase 11, and Phase 13's carrier quote feeds Phase 14. Phase 17 starts when each track has reached its last gate.

Phase 10: Speed and voice acceptance

Why first: a friend on a call answers in well under a second. Today the median is about 2.6 to 2.8 s, mostly model time (first token about 1.7 to 2.0 s). Nothing in the human-feel phase will land until this is fixed.

Gate 10: on the real stack, the median from end of speech to first audio is under 1.0 s and the slow end (p95) under 1.8 s, the 8 sales scenarios still pass, and Virat signs off the human microphone test.

- **WP-10.1 Restore real audio.** Virat tops up Cartesia credits. Claude Code re-runs the Test Call, proves a successful audio retry and the "audio played" state against the real service, and removes that limitation from `docs/02-voice/voice-runtime.md`. Done when: a real retry speaks and the transcript shows played, not just generated.
- **WP-10.2 Model bake-off.** With a paid key, run `scripts/voice_benchmark.py` and `python -m aurevia.evaluation.run` for each candidate (pinned names only: a Flash-Lite class model, the current Flash model, a Claude Haiku class model). Record time to first token, end-to-end, cost per scenario and style metrics. Done when: a default model and a fallback are chosen from numbers saved in `docs/02-voice/benchmarks/`.
- **WP-10.3 Cut model time.** Shrink the prompt (about 4,000 input tokens per turn today), use prompt caching where the provider supports it, and cut tool rounds: when the call reaches the next-step stage, pre-fetch free slots so booking takes one round, not two to three. Done when: model first token under 600 ms at the median and tool rounds per booking drop, with `tests/test_conversation.py` and the honesty rules unchanged.
- **WP-10.4 Turn-handling benchmark.** In the simulator, compare fixed endpointing (0.5 to 1.5 s) against dynamic endpointing and the turn-detector model, with preemptive text-to-speech on and off, and Deepgram key terms for company and agent names. Five runs per setting, then ten human calls for the top two. Change defaults only on evidence. Done when: the chosen settings and their numbers are written into the voice audit doc.
- **WP-10.5 Human acceptance.** Run `docs/06-testing/voice-manual-acceptance.md`: five browser calls (English and Hinglish) scored with the call-review rubric in the pilot runbook. Proposed pass mark: average 4 out of 5 on naturalness and timing, and no call with a safety failure. Done when: Virat records the decision in the doc.

Network distance (India-region hosting) is measured again in Phase 14.

Phase 11: Human feel

Goal: the call feels like two friends talking: the agent listens, reacts in short natural sounds, pauses at the right moments, and its voice carries warmth. It still says it is an AI when asked (rule 8).

Gate 11: in a blind listening test, at least 70% of ten listeners say they would happily stay on the call, and average scores are at least 4 out of 5 for naturalness, warmth, listening and timing. These pass marks are proposed; Virat confirms them.

Today: the prompt bans filler and stock openers, and the agent only recognises the prospect's "mm-hmm". It never says one itself. There is no emotion control.

- **WP-11.1 Backchannel spike (design first, then stop for approval).** Find how to play a short clip while the prospect is speaking without taking the turn or triggering barge-in, using the installed LiveKit Agents version. Options: a guarded `session.say`, or a second audio source mixed into the room. Write `docs/02-voice/backchannel-design.md`. Done when: the simulator shows a clip playing mid-prospect-speech with the prospect's turn intact.
- **WP-11.2 Backchannel engine.** A small module in the voice worker (`backchannel.py`) with these rules:
  - speak only after the prospect has talked for a few seconds, at a short natural pause inside their turn
  - a maximum rate (default about one per 8 seconds, set per agent: off, low, medium)
  - never during numbers, emails, phone numbers, a question to the agent, or right after the agent spoke
  - pre-rendered clips per voice and language, so each use costs nothing (English: hmm, okay, right, yeah; Hindi: haan, achha, theek hai, ji)
  - stop immediately if the prospect starts a new thought over it

  Done when: unit tests cover every rule, the simulator rate stays inside the configured range, and overlap incidents are counted and low.
- **WP-11.3 Agent-side reactions.** Raise the prompt version and allow short varied reactions ("oh nice", "hm, got it") tied to the prospect's mood, while keeping two short sentences, no echoing of the prospect, and one question at most. Update `evaluation/style.py` so the gate rewards variety instead of banning every opener. Done when: 8 of 8 scenarios pass and word counts stay near 18 per reply.
- **WP-11.4 Mood and emotion.** Extend the background bookkeeping pass to record `prospect_mood` (warm, neutral, rushed, annoyed, skeptical, confused). Map it per turn to speed and emotion through the Cartesia adapter (`update_options`): softer and slower when annoyed, lighter when warm. Text-based first; voice-tone analysis only later if needed. Done when: simulator personas produce different voice settings, the setting is stored in turn metrics, and tests use a mocked provider.

- **WP-11.5 Natural speech.** Say rupee amounts, dates and phone numbers the way people do. Add a pronunciation dictionary for company and product names. Switch language per turn for Hinglish (the language switch exists) and benchmark it on mixed-language audio before turning it on.
- **WP-11.6 Turn-taking polish.** Handle: a polite "sorry, go ahead" after being interrupted, a prospect saying "hold on", "can you repeat that", "I'm driving", long silences, and a clean goodbye. Reuse the existing silence tracker and interruption handling.
- **WP-11.7 Human-feel scorecard.** Blind A/B listening test of builds with and without each feature, plus numbers compared with the expert callers (from Phase 12): backchannels per minute, pause length before replying, talk ratio. Done when: Gate 11 is met and the scorecard is saved in `docs/06-testing/`.

Phase 12: Learning from the expert recordings

Goal: turn Virat's 50 to 60 recordings of top cold callers into two things: an approved playbook of what to say (objections, hooks, closes), and a timing profile of how they say it (pauses, backchannels, talk ratio). This follows `docs/07-learning/sales-learning-design.md`, which today is design only, and adds the audio step it leaves out.

Gate 12: playbook v1 for the first niche is approved by a person, a simulated A/B shows better objection handling and booking than the no-playbook baseline with zero honesty or compliance regressions, and the timing profile feeds Phase 11.

Principles carried over: authorized sources only; personal data redacted first; correlation is not proof (only simulation and A/B count); a person approves every technique; techniques never bypass the sales-state engine, tools, compliance gate or AI disclosure; no fine-tuning of any model.

- **WP-12.0 Inventory (Virat).** One spreadsheet row per recording: file, caller, proof of the caller's permission, language, industry and product, outcome (booked or not), length, and any evidence about the prospect's consent to being recorded. Pick the first niche from this. Done when: the sheet is complete and counsel has seen the prospect-consent question.
- **WP-12.1 Transcribe with timing.** A batch speech-to-text adapter behind the provider layer, with speaker separation and word-level timestamps, covering English, Hindi and Hinglish. Audio is used only to make the transcript and is then deleted; encrypted storage in an Indian region, per-tenant keys and audited access are required before any audio is stored (design section 6). Done when: three calls are checked by a human and word error rate is recorded.

- **WP-12.2 Redact.** Remove names, phone numbers, emails, addresses and ID numbers with patterns plus a model detector through the Model Gateway. Placeholders keep the structure. Delete the original after success. Done when: a report shows counts only, never values, and a test seeded with fake personal data finds none left.
- **WP-12.3 Tables, permissions, import.** New migrations (never edit applied ones) for `learning_sources`, `learning_transcripts`, `learning_labels`, each with `tenant_id` and forced row-level security; permissions `learning.import`, `learning.review` and `learning.publish`; CSV and JSON import. Done when: tenant-isolation tests pass for every new table.
- **WP-12.4 Label.** For each call: stage markers (opener, hook, discovery, pitch, objection, close), objection category, outcome, qualification facts, next step, talk ratio, question count. The model proposes, and a person spot-checks about one in five. Done when: label accuracy against the human check is recorded.
- **WP-12.5 Timing profile.** From word timestamps compute, per stage: pause before replying (p10, p50, p90), overlap rate, backchannels per minute and where they fall, speech rate, turn length. Store aggregates only. No voice fingerprints. Done when: a profile file exists and Phase 11 reads it for its defaults.
- **WP-12.6 Technique extraction and review.** Candidate techniques with redacted excerpts, how often they appear in booked and unbooked calls, and conditions (stage, objection, language). An automatic policy check runs first (honesty, invented facts, pressure tactics, compliance). A dashboard review queue lets a person approve, edit or reject, and every decision is audited.
- **WP-12.7 Playbooks and retrieval.** Immutable versions (draft, active, retired), one active per agent. Each turn retrieves at most three matching techniques and adds them to the prompt as guidance below the honesty rules. Retrieval adds under 50 ms and records which techniques were offered.
- **WP-12.8 Evaluate.** Simulator A/B: baseline (no playbook) against the candidate, several runs on scripted prospects plus new ones built from objections seen in the recordings. Only an improvement with no honesty or compliance regression can become active. Then a live A/B on 50% of pilot calls, with the version stored per call.

Voice cloning is out of scope. Cloning any caller's voice needs that person's separate written consent for cloning and a counsel check. The default is a licensed catalogue voice (Meera or Dev).

Phase 12 addition: sources and the seed playbook

The playbook is built from two kinds of material, and they are used differently. **Real calls** (audio plus a transcript with speaker labels) feed both the techniques and the timing profile. **Talks and interviews** about cold calling (transcript only, or a voice recording made from one) feed techniques only, because a studio conversation says nothing true about phone timing. The pipeline must accept both types and must never build a timing profile from a talk.

Work packages

- Add a `source_type` field (`call` or `talk`) to the learning pipeline. A `call` needs speaker labels; a `talk` skips the timing step.
- Write playbook entries in one fixed format: situation, what the agent says, tone, goal, source, and status (`proposed` or `approved`). Nothing is used in a call until Virat approves it.
- Seed the first 15 to 20 entries from the material below, then test them in practice calls (Gate 12).

Call structure (from the expert material; test every step)

1. Open warmly with the first name only and a short question, never a pitch. The agent does not pretend to know the person and says it is an AI when asked.
2. When they ask who it is, give only first name and company, then ask "is this a good time?" early, so that "I am busy" is handled before the pitch.
3. One sentence of value: what we do, through what, to get what outcome.
4. Check interest with a question, then ask for the meeting.
5. End every turn with a question, not a statement.

Goal ladder. Try for a meeting first. If declined, ask permission to stay in touch or call back. If declined, ask who else it might suit (a referral). Only then end politely. Before accepting "send me an email", ask for the meeting once. A clear "do not call me again" overrides the whole ladder: stop at once, record it, and never contact that number again.

Objection method. Agree first ("I completely understand"), then ask why, then isolate ("is that the only thing stopping you?"), then act: realign, educate or disqualify. Separate "no money" (disqualify) from "money allocated elsewhere" (realign). A useful probe is a 1 to 10 question: "why not a 1?" and "what would make it a 10?". Log each objection with its type (price, timing, authority, need) using the existing `log_objection` tool.

Booking. Offer two choices, not an open question ("Thursday or Friday?", then two times). Send the invite while the person is on the line, confirm it shows on their calendar, and schedule a reminder before the meeting.

Qualification. Check budget, authority, need and timeline. On a cold call the aim is to disqualify early, so the meeting is only booked with people who can buy. Each lead ends as qualified, disqualified, educate (keep warm) or realign.

Tone by stage (maps to the existing mood and voice control events; amounts are starting points to tune by ear)

Stage	Tone
Opening	Friendly, medium pitch and volume
Discovery	Calm, lower pitch, slower
When they share a problem	Empathetic, slow, short acknowledgement
Objection	Start low and soft, then firmer; pause after confirming
Price	Confident and steady, never apologetic
Creating urgency	Brighter and faster, used rarely

Calls that never reach a person (lands in Phase 14). In one sample of 50 dials, only about 3 became real conversations, so detection speed drives cost. Detect voicemail, phone menus, receptionists and wrong numbers in the first seconds (table in the System Design).

Calling windows and retries (lands in Phase 15). Call inside the prospect's local time window, with 3 to 5 pm as the first window to test, and always inside the compliance gate's legal hours, which win any conflict. Retry a no-answer on another day and another time, honour "call me in a week", and stop after a set number of attempts. Wrong numbers leave the sequence.

Gate 12 addition: at least 10 seeded entries approved by Virat, and a test call where the agent follows the goal ladder and tone changes by stage.

Phase 13: Make it cheap

Goal: the sales pitch is "cheaper than your 10 or 20 person team", so variable cost must fall under 2 rupees per talk minute, and cost per booked meeting must be visible for every customer.

Gate 13: a measured (not estimated) cost of 2 rupees or less per talk minute across at least 20 real test calls, a cost-per-booked-meeting report per tenant, and an ROI calculator that states its assumptions.

Today's estimate (about 90 rupees to the dollar; confirm the current rate; hosting not included):

Part	Estimate per minute	Main lever
Speech-to-text	about 0.7 rupees	cheaper plan, multilingual mode, self-hosting at volume
Text-to-speech	1.2 to 2.1 rupees	pre-rendered clips, cheaper voice, volume plan
AI model	0.45 to 1.35 rupees	light model by default, prompt caching
Phone line	0.4 to 0.6 rupees	compare carriers, per-second billing
Total	**about 2.7 to 4.7 rupees**	

Sources: Deepgram streaming rate, Cartesia plans, Plivo India rate, and the model cost in `docs/02-voice/benchmarks/`. Exotel does not publish a per-minute rate, and bills in 60-second pulses.

- **WP-13.1 True cost metering.** Check that the usage engine (`usage/costs.py`, `usage/recorder.py`, `docs/03-platform/cost-and-usage-engine.md`) records speech-to-text seconds, text-to-speech characters, model tokens (cached and uncached), phone minutes and pulses. Prices live in a versioned config, never in code. Report cost per call, per minute, per booked meeting and per tenant. Add a budget alert and a per-tenant kill switch. Done when: 20 test calls give a real cost per minute, saved in the docs.
- **WP-13.2 Voice bake-off.** Blind listening plus cost for the current voice, a volume plan, other providers including Indian-language specialists, and a self-hosted open model at the volume where it breaks even. Choose by quality per rupee. The adapter layer stays.

- **WP-13.3 Audio cache.** Pre-render the greeting, backchannels, closing lines and common phrases per voice and language, keyed by text, voice, language and emotion. Track the hit rate. This also cuts latency.
- **WP-13.4 Model routing.** A small fast model for routine turns, a stronger one only for objections, pricing and booking. Prompt caching, stage-aware context trimming, and summaries for long calls. Done when: the 8 scenarios still pass and cost per scenario drops.
- **WP-13.5 Do not pay for silence.** Start speech services only after the call is answered. Detect voicemail and answering machines and end or handle them per policy. Cap call length per campaign. Skip the model for very short non-answers. Keep the existing silence cut-off.
- **WP-13.6 Phone rates.** Get written quotes from Plivo, Exotel and at least one more India carrier for outbound to mobile at volume. Compare per-second against per-minute billing. Choose behind the telephony adapter.
- **WP-13.7 Pricing and ROI.** Plans in the platform admin: a price per booked meeting plus a per-minute floor, with the gross margin per tenant shown to Virat. An ROI calculator takes the customer's team size, salaries and talk time and compares their cost with Aurevia's.

Be honest in the calculator. Pay data is thin: telesales executives average about 2 lakh a year and SDRs about 9.8 lakh, both from small and old samples. Against SDR-level teams Aurevia is far cheaper. Against entry-level telecallers the margin is thin at 3 to 5 rupees a minute, which is why the under-2-rupee target matters.

Phase 14: Phone go-live

Goal: real calls to real phones, hosted in India, inside the law, with the safety rails already in the code left intact.

Gate 14: staging runs on an India-region server with HTTPS; a real phone call to Virat's own registered number works end to end with phone-audio quality measured; a real do-not-disturb registry is integrated; counsel's review of the exact policy version is recorded; backups are restored successfully at least once; and the pilot acceptance matrix passes.

Start by reading: `docs/02-voice/telephony.md`, `docs/05-operations/hosting-plan.md`, `cloud-setup.md`, `deployment-and-environments.md`, `disaster-recovery.md`, `pilot-runbook.md`, and `docs/04-compliance-security/counsel-review-pack.md`.

Known state: hosting Option A is decided (one server with Docker Compose and Caddy, in an India region), but provider, domain and budget are open. The Exotel trunk is untested. `providers/dnd.py` returns "unknown" for every number and the policy blocks unknown, so live calling is impossible until a real registry is connected. The counsel pack says the first customers are insurance brokers.

- **WP-14.1 Choose and build staging.** Virat picks provider, domain and budget. Claude Code builds the server as the hosting plan describes: Caddy HTTPS, private Postgres and Redis, real LiveKit keys, firewall rules, a root-only secrets file, the dashboard built with the public API address. Done when: the health endpoint is ready on the public address and a browser Test Call works from outside the home network.
- **WP-14.2 Measure region and media options.** Run the benchmark from the India server and from one region nearer the AI providers. If the extra delay or the UDP and NAT work is painful, price LiveKit Cloud managed media as the hosting plan's Option C. Done when: the decision and numbers are written into the hosting plan.
- **WP-14.3 Phone trunk.** Connect LiveKit SIP to the carrier chosen in WP-13.6, open the SIP ports to the carrier's addresses only, register the caller ID numbers. Done when: in test mode the system calls Virat's own phone and an inbound call reaches the right tenant.
- **WP-14.4 Phone-audio quality.** Phone audio is narrow-band and noisy. Pick the speech model setting that works best on it, then measure word error rate on real phone calls in English and Hinglish, and check barge-in and echo on a real network. Done when: the numbers are in the voice audit doc. Synthetic audio does not count.
- **WP-14.5 Real do-not-disturb registry.** Integrate the national registry lookup (through the carrier or an approved scrubbing service) behind the existing `DndRegistry` interface, with caching rules from counsel. Unknown must still mean blocked. Done when: tests cover registered, not registered, unknown and a failing lookup.
- **WP-14.6 Registrations (Virat and counsel).** Settle who registers on the telecom DLT platform, the correct caller ID series for sales and service calls, carrier know-your-customer, and the sender and number approvals. Record the answers in the policy pack questions.
- **WP-14.7 Counsel review.** Give the review pack to Indian counsel. Turn every change they ask for into a new immutable policy version, then record the review with `python -m aurevia.compliance.operator review VERSION --by ... --reference ...`. Ask counsel specifically about: whether the agent must say it is an AI at the start of every call, recording and transcript notice, calling hours and holidays, consent wording, and insurance rules. Done when: the review is recorded against the exact version used.

- **WP-14.8 Call safety.** Voicemail and answering-machine handling, pacing and concurrency limits per tenant, an abandoned-call limit, daily caps, an instant campaign kill switch, retry rules, and in-call opt-out already handled by the do-not-call tool. Done when: tests prove each limit and the kill switch.
- **WP-14.9 Operations.** Alerts for failed calls, provider outages and cost spikes; scheduled backups with a restore check; an incident runbook; Virat as the named responder. Run `pilot-acceptance-tests.md` and fill the acceptance matrix.

Live mode stays off until WP-14.5 and WP-14.7 are done.

Phase 15: Integrations that replace the sales workflow

Goal: the agent finishes the job without a person: the meeting lands on a real calendar, the call result lands in the customer's CRM, and the follow-up goes out on the right channel.

Gate 15: a meeting booked on a call appears on the customer's real calendar with an invite; the outcome, notes and summary appear in their CRM; a follow-up message is sent only after a consent check; a no-show is recovered in a test run.

Today: bookings are internal records (`sales/scheduling.py` notes that Google and Outlook can sit behind the same tool interface), and the agent can only schedule a follow-up that a colleague sends.

Build each integration behind the adapter layer. Every tool result must still say `ok` truthfully, so the agent never claims a booking or message that failed.

- **WP-15.1 Calendar.** Google Calendar first, Outlook second. Per-tenant sign-in, tokens encrypted at rest, free and busy times turned into slots, events created and cancelled with an invite and meeting link, time zones handled. Keep the unique index that blocks double booking. Done when: tests against a sandbox calendar cover races, a revoked sign-in and an outage, and each failure returns a not-ok tool result.
- **WP-15.2 CRM.** HubSpot and Zoho first (Zoho is common in India), Salesforce later. Import leads; push outcome, stage, qualification facts, notes, transcript summary and meeting. Per-tenant field mapping, duplicate matching, safe retries with idempotency, and visible sync status. A generic signed webhook and CSV export cover everything else.
- **WP-15.3 Messaging follow-ups.** A sender for WhatsApp Business (approved templates only, opt-in recorded), SMS (templates registered on DLT) and email (SPF, DKIM, DMARC, unsubscribe link). A validated, audited tool sends only after the compliance

gate checks consent for that channel and purpose, inside allowed hours. Delivery status is tracked and opt-outs are honoured at once. The model chooses a template and fills its fields; it does not write free-form messages.

- **WP-15.4 After the call.** Within seconds of hang-up: a summary, next steps and a lead score. Follow-up sequences by outcome (booked: confirmation, then reminders a day and an hour before; interested but not booked: a message and a retry; no answer: retry within policy). No-show recovery with a reminder and a reschedule flow.
- **WP-15.5 Handoff and callbacks.** A warm handoff alert to the right colleague (email, WhatsApp or Slack) with the call summary, building on the existing handoff tool. A prospect who says "call me at 5" is queued as a scheduled callback.
- **WP-15.6 Public API and webhooks.** Per-tenant API keys with scopes and rate limits, signed outgoing webhooks (`call.completed`, `meeting.booked`, `lead.updated`), and API docs generated from the OpenAPI file, so tools like Zapier or Make can connect.

Phase 16: A product companies can buy

Goal: a company signs up, teaches the agent its business, tests it, launches a campaign and sees results and cost, without Virat in the room.

Gate 16: a new test company finishes onboarding unaided in under an hour (proposed target), launches a test-mode campaign, sees cost and results, and receives a correct invoice for its test usage.

Today: the dashboard already has agent setup, test call, leads, campaigns, calls with transcripts, compliance, audit, usage, team and platform admin, plus an onboarding page. This phase fills the gaps between those pieces.

- **WP-16.1 Onboarding wizard.** One guided path: company details, knowledge base, agent personality and voice, qualification questions and objections, a test call, connect calendar and CRM, import leads, launch in test mode. Progress is saved and every step says what is missing.
- **WP-16.2 Knowledge base.** Upload documents and web pages, or type FAQs and prices. Chunk and retrieve per tenant so the agent states product facts only from what the company supplied; when it does not know, it offers a colleague follow-up. Retrieval must fit the latency budget (together with playbook retrieval, under 100 ms) and the answer source is visible in call review.

- **WP-16.3 Vertical templates.** A ready agent per industry (insurance first, because the counsel pack names insurance brokers): personality, qualification questions, objection guidance and escalation rules. Templates are starting points the company edits.
- **WP-16.4 Try it before launch.** Practice against simulated prospects (friendly, busy, skeptical, price-sensitive, confused, strong-objection) from the dashboard and see the transcript, the stage path and any problems found.
- **WP-16.5 Call review.** Transcript with tool actions, stage changes, mood, techniques used and knowledge sources; tags, a quality score and a thumbs up or down that feeds the review queue. Audio playback only if a later, consented recording feature exists.
- **WP-16.6 Analytics.** The funnel (dialed, answered, talked, qualified, booked, attended), cost per booked meeting, best times to call, objection counts, and A/B results for playbooks.
- **WP-16.7 Billing.** Razorpay first (Stripe if selling abroad), GST invoices, usage-based charges from the usage engine, trial credits, plan limits (already enforced), failed-payment handling and a suspend-and-resume path that uses the existing platform admin.
- **WP-16.8 Account safety.** Two-factor sign-in for owners and admins if not already present, session controls, and an invitation flow check. SSO is later.
- **WP-16.9 Owner view for Virat.** Platform admin shows per-customer cost, revenue and margin, call health, provider status and the kill switches from Phase 14.
- **WP-16.10 Customer help.** In-app help for each setup step, a status page, a support email, and the terms of service, privacy policy and data processing agreement that counsel approves (see Phase 17).

Phase 17: Pilots and launch

Goal: prove the agent with two or three real companies, with numbers a buyer believes, then start selling.

Gate 17: two or three pilots finish four weeks; each has a measured cost per booked meeting next to the company's own team baseline; at least one company agrees to pay or to be a named case study; the legal pack is in place.

Most of this phase is Virat's work. Claude Code supports it with the tooling below.

- **WP-17.1 Pick the niche and the ideal customer.** Candidates: insurance brokers (the counsel pack's first customer) or the industry of the expert recordings, whichever matches. Criteria: many outbound calls today, a meeting worth real money, an existing

human team to compare against, India-based, willing to share results.

- **WP-17.2 Design the pilot.** Four weeks. Agree the success numbers in writing before starting: connect rate, qualified leads and meetings per 100 dials, meetings attended, and cost per booked meeting, each compared with the company's own callers on similar leads. Start with warm leads (people who enquired), because consent is clearest there. No live outbound before Gate 14.
- **WP-17.3 Legal pack (counsel).** Pilot agreement, terms of service, privacy policy, data processing agreement (the customer as data fiduciary, Aurevia as processor, as the counsel pack assumes), and acceptable-use rules (no calling numbers on the do-not-disturb list, no deceptive scripts). Decide which legal entity contracts with customers and issues GST invoices.
- **WP-17.4 Sales material.** The ROI calculator from WP-13.7; a landing page; a two-minute demo video; a "request a demo call" form that records the visitor's express consent and then calls only that person through the compliance gate; a compliance one-pager (AI disclosure, do-not-disturb handling, audit trail, data deletion); a case-study template.
- **WP-17.5 Run the pilot.** Daily call review in week one, a fix list, an automatic weekly report per pilot company (funnel, cost per meeting, best objections, problems found), and the incident process from Phase 14.
- **WP-17.6 Decide.** If cost per booked meeting beats the human baseline and the quality scores hold, finalise pricing and sell. If not, return to Phase 13 or Phase 11 before adding customers.

What Claude Code builds here: the weekly report generator, the demo-call form with consent capture, pilot tenants with caps, and a pilot results page. Everything else is business work for Virat and counsel.

Phase 18: Later, after Gate 17

Do not start any of this early (rule 10). Each item has a trigger that says when it becomes worth building.

- **Marketing operations (stage 3 of the roadmap).** Compliant lead sourcing and list hygiene, campaign planning, script A/B testing at scale, attribution. *Trigger:* paying customers ask for help finding leads, not only calling them.
- **Marketing content and outreach (stage 4).** Email and WhatsApp campaign drafts, landing pages, ad copy, all through the same consent and template controls. *Trigger:* sales calling is stable and profitable.

- **More languages.** Tamil, Telugu, Marathi, Bengali, Gujarati and others, each with its own voice test, speech-recognition error check and recordings-based playbook. *Trigger:* a signed customer in that language.
- **Inbound receptionist and support agent.** Answer inbound calls, qualify, book. The inbound path already exists. *Trigger:* customers ask for it.
- **Scale and reliability.** Split services (hosting plan Options B or D), autoscaling workers, multi-region media, standby database. *Trigger:* sustained load that one server cannot carry, or a customer's availability requirement.
- **Lower cost at volume.** Self-hosted speech-to-text and text-to-speech on rented GPUs when monthly minutes pass the break-even found in WP-13.2.
- **Voice-tone emotion detection.** Read the prospect's tone from audio, not only their words. *Trigger:* text-based mood (WP-11.4) shows a clear ceiling.
- **Consented call recording and playback.** Only with the design's section 6 requirements met: notice, stored consent, fixed retention, encrypted Indian storage, separate approval.
- **Enterprise features.** Single sign-on, per-team permissions, audit exports, a service-level agreement.
- **Other countries.** A new policy pack per country (for example United States and United Kingdom) with its own counsel review before any live call.
- **Reseller and white-label.** Agencies selling Aurevia under their own brand.

Cross-cutting work (runs through every phase)

Security (builds on `docs/04-compliance-security/`)

- Add dependency and container scanning and secret scanning to CI (Python, npm and Docker images).
- Treat every imported transcript, uploaded knowledge document and prospect utterance as untrusted data. Add prompt-injection and tool-abuse tests for each new input path.
- Run the cross-tenant isolation test for every new table. Verify the audit chain on a schedule.
- Rotate all provider keys before the first pilot, use least-privilege access on the server, and get an independent penetration test before the first paying customer.

Reliability

- Add a second speech-to-text and a second text-to-speech provider behind the adapters, with circuit breakers like the model gateway already has. On a phone call a prospect is never left in silence: say a short line, then end politely if the provider stays down.
- Set targets and watch them: call setup success rate, median and p95 reply time, uptime.
- Load-test concurrent calls (start with 20, then 100) and record cost and quality at each level. Test provider failures on purpose.
- Follow `disaster-recovery.md`: written recovery time and data-loss limits, a restore drill every quarter.

Testing and evaluation

- Keep the evaluation harness as the gate. Add Hinglish scenarios, the eight personas, an objection set built from the recordings, and adversarial cases: "are you a human?", prompt injection, abusive prospects, wrong number, do-not-call request, a child or elderly person answering, a request to speak to a manager.
- Every pilot failure becomes a regression test.
- A nightly evaluation run with a cost cap. Reports always separate mocked tests, real-provider tests, synthetic audio and human calls.

Observability

- One trace id per call across worker, backend and providers. Structured logs without personal data or secrets.
- Dashboards for time at each stage, error rate by provider, cost per minute, and call outcomes, with alerts on spikes.

Data protection (India's DPDP Act)

- Keep consent records, 90-day transcript retention, export and erasure working, and add a grievance contact and a breach-response plan.
- Ask counsel about sending personal data to speech and model providers outside India, and publish a sub-processor list. Redact personal data from prompts where the call does not need it.

Documentation and decisions

- After every phase, update `CLAUDE.md` status, the phase docs and the acceptance matrix. Record each major decision (provider, hosting, pricing) in the docs with its date and the numbers behind it.

Decisions and tasks only Virat can do

Claude Code cannot do these. Each one blocks work in the phase named, so do them early.

Task or decision	Who helps	Needed before
Paid model key; top up Cartesia and Deepgram credits; set a monthly provider budget	none	Phase 10
Confirm the pass marks for speed and human feel	none	Phases 10 and 11
Listen and choose: Hindi and Hinglish voice, how often the agent says "hmm" and "okay"	ten test listeners	Phase 11
Recordings inventory sheet: caller permission on file, language, industry, outcome, prospect-consent evidence	callers	Phase 12
Pick the first niche (insurance brokers, or the recordings' industry)	none	Phases 12 and 17
Voice cloning: yes or no (the default in this plan is no)	each speaker, counsel	Phase 12
Hosting provider, domain and budget	none	Phase 14
Choose the phone carrier after written quotes; supply business documents for carrier verification	carriers	Phases 13 and 14
Telecom registrations (DLT, caller ID series)	counsel, carrier	Phase 14
Hire Indian counsel; answer their questions; get the written opinion	counsel	Phase 14
Legal entity for contracts and GST invoices; bank account; payment gateway account	accountant	Phases 16 and 17
Final pricing	none	Phases 13 and 17
Find two or three pilot companies and sign pilot agreements	counsel	Phase 17

Task or decision	Who helps	Needed before
Review and merge every pull request (Claude Code never merges)	none	every phase

Risks and the fallback for each

Risk	Fallback
Replies stay slower than 1.5 s	Faster paid model, preemptive speech, audio cache, closer region or managed LiveKit media, shorter replies
Cost stays above 2 rupees a minute	Price per booked meeting, target teams that employ costly SDRs, cheaper voice, self-hosted speech at volume
Counsel requires stricter consent or an AI notice at the start of every call	Put the notice in the opening line, issue a new policy version, focus first on warm leads who enquired
Rules change and promotional AI calling is restricted	Concentrate on service calls, inbound and follow-ups; stay in test mode until review
Backchannels and emotion sound fake or annoying	Conservative defaults, per-agent off switch, blind listening tests before any default changes
The recordings are too few or too narrow	Add objections from simulation, grow the library from pilot calls with opt-in, start with one niche
A speech, voice or model provider fails or raises prices	Adapters plus a second provider per layer, circuit breakers, spend alerts
Caller ID is labelled as spam	Properly registered numbers, a low abandoned-call rate, honest scripts, reputation monitoring
A pilot shows no gain over human callers	Return to Phases 11 and 13, or switch the first product to inbound answering and follow-ups

Risk	Fallback
One person carries the whole build	Keep phase order and gates, one pull request per work package, use counsel and a sales partner for the non-code work

Launch checklist: what complete means

The product is complete when every box is ticked. Claude Code reports against this list at the end of each phase.

Product

- [ ] Median reply under 1.0 s and p95 under 1.8 s on real phone calls
- [ ] Human-feel scorecard passed (Gate 11)
- [ ] Playbook v1 active for the first niche, with a simulated and live A/B showing a gain and no honesty or compliance regression
- [ ] Measured cost at or under 2 rupees per talk minute; cost per booked meeting reported per customer
- [ ] Meetings go onto the customer's real calendar and outcomes into their CRM
- [ ] Follow-ups are sent on WhatsApp, SMS or email only after a consent check
- [ ] A new company onboards unaided and is invoiced correctly

Safety and legal

- [ ] Real do-not-disturb registry connected; unknown still blocks
- [ ] Counsel's review recorded against the exact live policy version
- [ ] Telecom registrations and caller IDs complete
- [ ] The agent says it is an AI when asked (and at the start of every call if counsel requires), proven by tests
- [ ] Tenant isolation and audit-chain checks pass; provider keys rotated; penetration test passed
- [ ] Backups restored successfully; alerts and the incident runbook in place

Business

- [ ] Two or three pilots finished with measured results, and at least one company paying or agreeing to be a case study
- [ ] Legal pack signed off; legal entity, GST and payment gateway ready

- [ ] Pricing page and ROI calculator live
- [ ] `CLAUDE.md`, the phase docs and the acceptance matrix are up to date and all green